

SNT

**Interdisciplinary
Centre for Security,
Reliability and Trust**

High Range Resolution Software Defined Radar

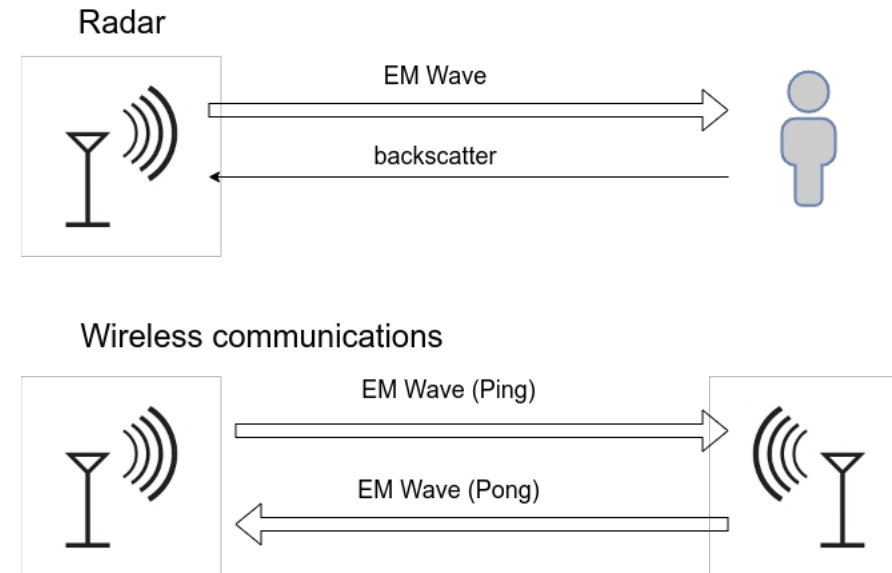
Motivation

Radar: Transmitting RF signals and then analysis of their reflections

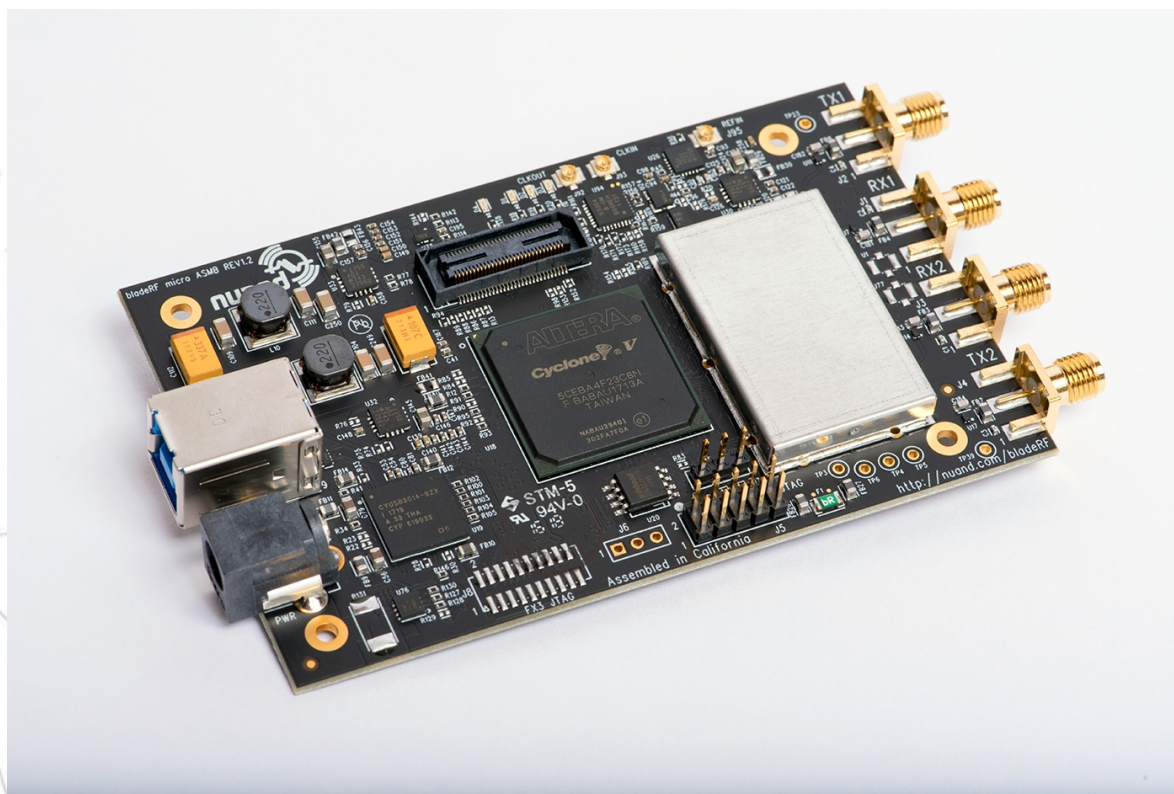
Software Defined Radio: a radio that can change its purpose with a simple software update

Software Defined Radar: Radar implemented on top of a SDR

- Flexibility
- Cost-Effectiveness
- Adaptability
- Rapid Prototyping



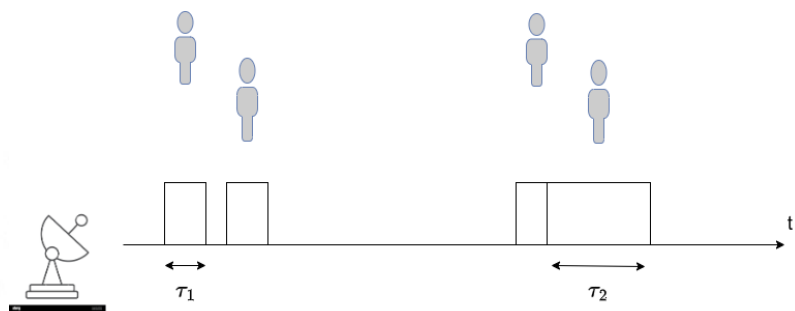
BladeRF 2.0 Micro



Spec	Value
ADC/DAC Sample Rate	61.44 MHz
Max IBW	56 MHz
RF Tuning Range (TX)	47 MHz to 6 GHz
RF Tuning Range (RX)	70 MHz to 6 GHz
CW Output Power	Around 8 dBm (6.3 mW)
Dimensions	6.3 x 10.2 x 1.8 cm ³
Weight	90 g
Host communication	USB 3.0

\$540

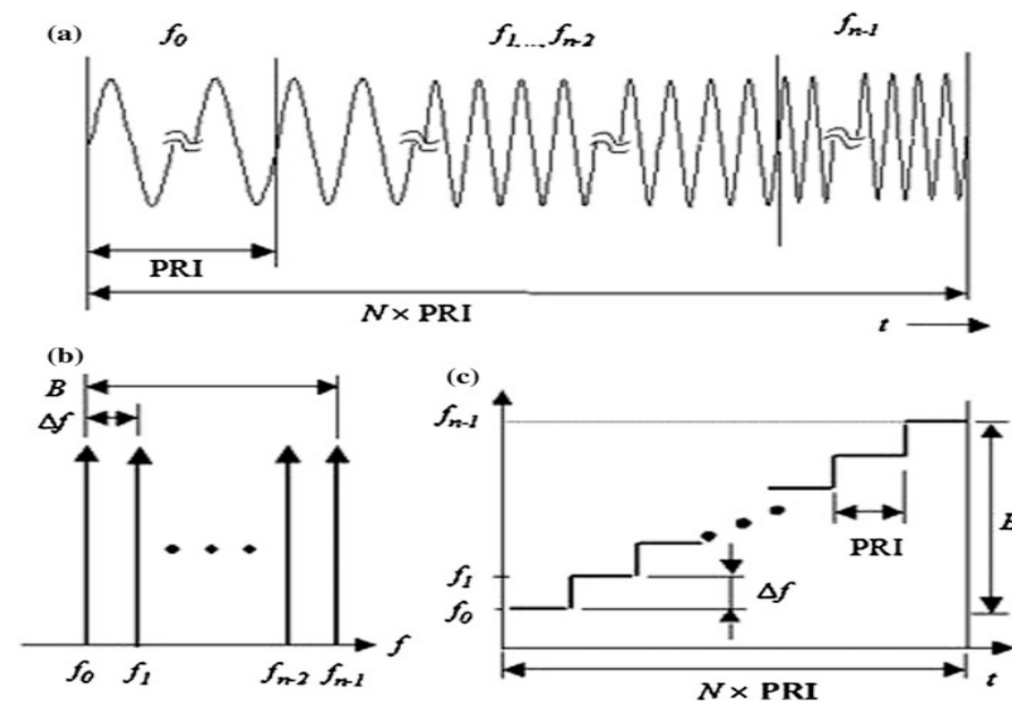
Challenge 1: Radar Range Resolution



- Shorter pulse → Larger bandwidth → Finer resolution
- Range Resolution = Speed of Light / (2 * Bandwidth)

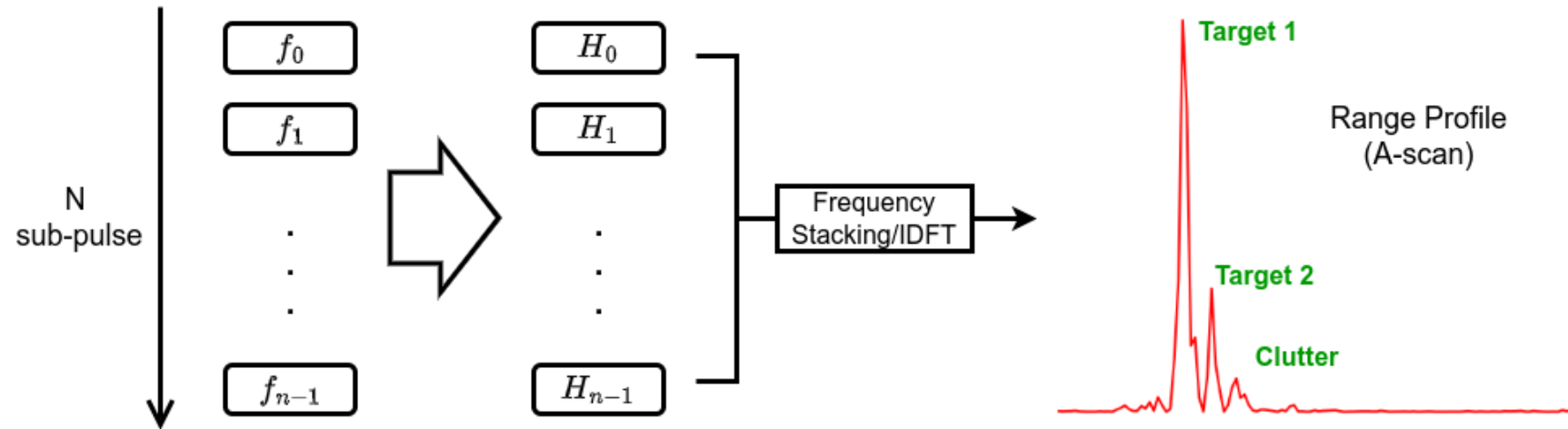
- **2 GHz → 0.075m**
- **1 GHz → 0.15m**
- **56 MHz → 2.73m**
- **10 MHz → 15m**

Type of Radar	Output Power	Instantaneous Bandwidth
Pulsed Radar	High	Wide
FMCW Radar	Low	Wide
SFCW Radar	Low	Narrow



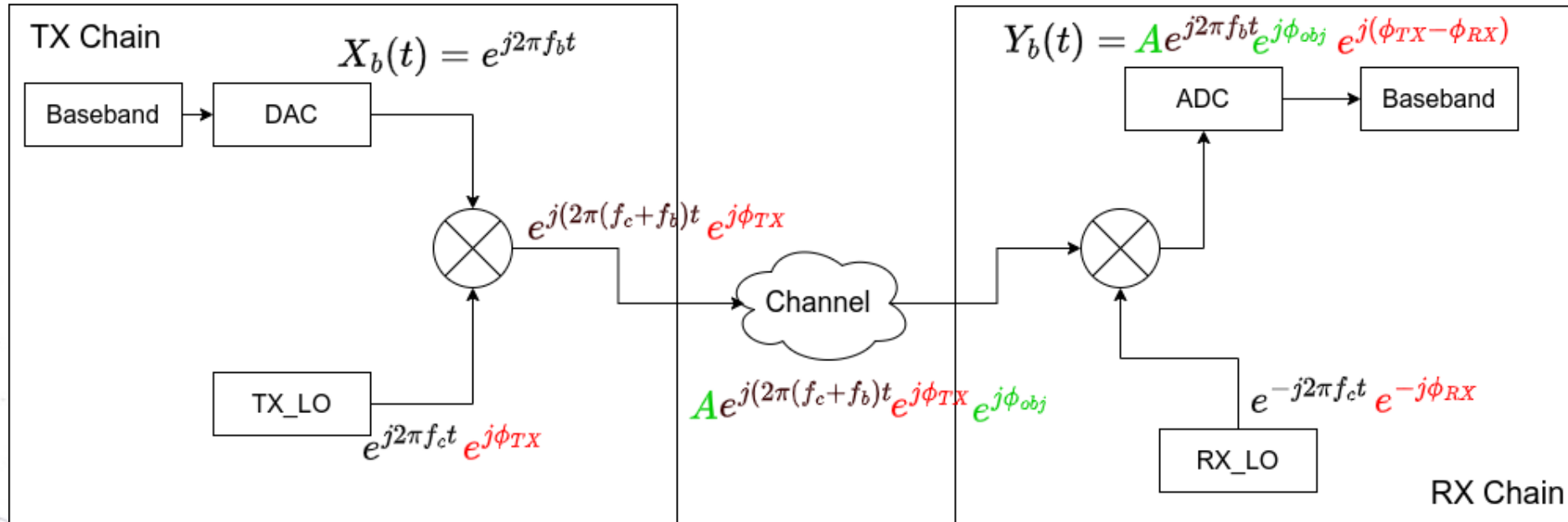
SFCW radar: a) waveform, b) frequencies and c) spectrogram

SFCW Radar



- Transmit pulse and receive echo at each frequency step
- Calculate channel responses for each frequency step
- Coherently process the collected channel responses to get range profile

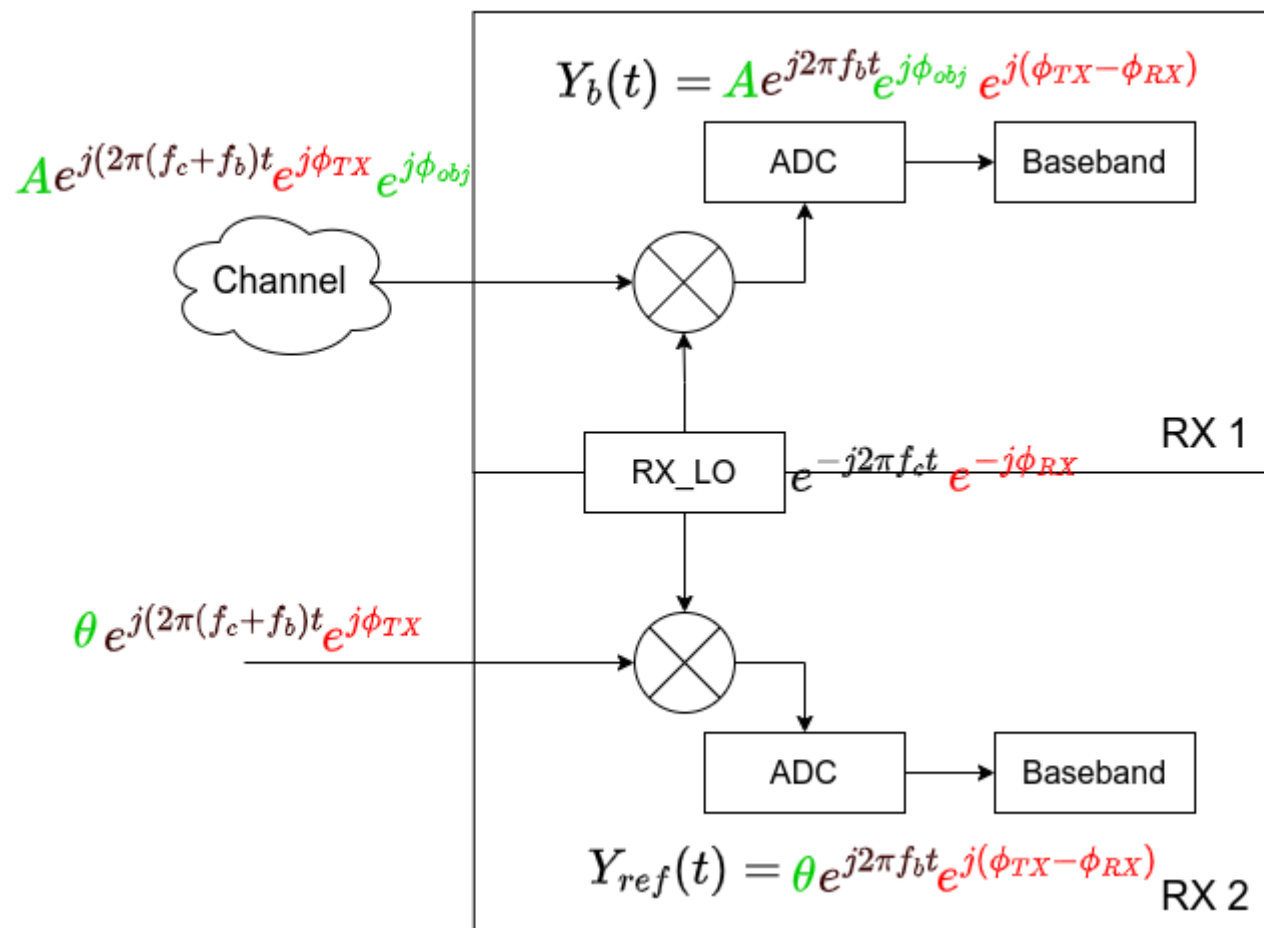
Challenge 2: None-coherent TX/RX



$$H_{f_c} = Y_b(t)/X_b(t) \Rightarrow A e^{j\phi_{obj}} \underbrace{e^{j(\phi_{TX}-\phi_{RX})}}$$

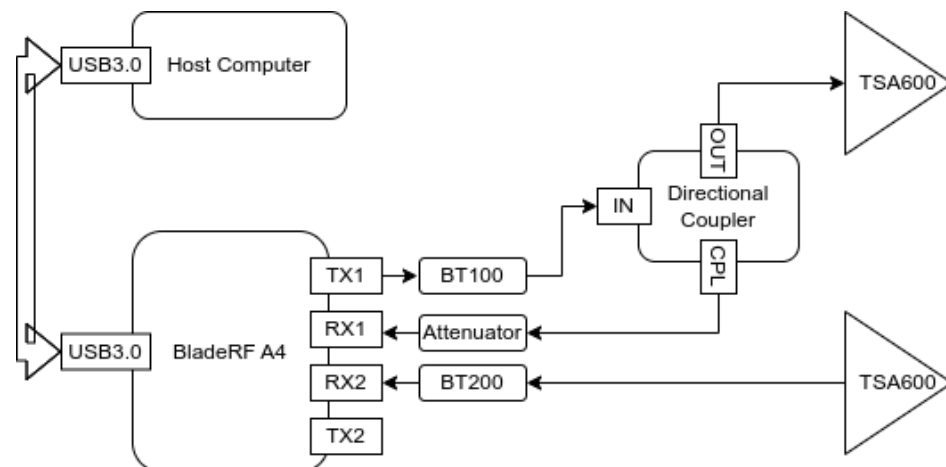
Changes randomly at each frequency step

Use MIMO



$$Y_b(t)/Y_{ref}(t) \Rightarrow Ae^{j\phi_{obj}}/\theta$$

A constant



Challenge 3: Internal Coupling

- The received signal from RX channels is the sum of radar echos, a copy of TX signals and noise due to internal circuit
- Internal coupling won't hurt much for communication purpose but introduce strong phase noise for radar application
- A calibration method was proposed to reduce the impact of internal coupling (See the upcoming paper)
- The method reduces the SCR from -7db to -14db

Results: A Low-cost Software Defined Radar

Real time re-configurable

- Real time re-configurable
- Build in single tone and chirp waveform, support user defined waveform
- Minimum range resolution: 3 cm
- Noise floor: -70 db
- Detection range: 30 meter for door size objects
 - 1 GHz effective bandwidth
 - Central frequency 2.4 GHz
- Penetrating capability: 0.5 meter
 - Dry sands
 - 20 cm x 20 cm reflection surface
 - 1 GHz to 2 GHz bandwidth

Hardware

BoM

Components	Units	Unit price	Spec
BladeRF micro A4	1	540 US dollar	Software defined radio with 2x2 MIMO channels. Frequency range: 47MHz to 6GHz Max sampling rate: 61.44MHz
BT-100 Bias-tee	1	30 US dollar	Power amplifier (TX) Frequency range: 50 MHz to 6 GHz Typ Gain: +15.6 dB at 200 MHz, +10.7 db at 2.4 GHz
BT-200 Bias-tee	1	30 US dollar	Low noise amplifier (RX) Frequency range: 50 MHz to 6 GHz Typ Gain: +20.2 db at 200 MHz, +16 db at 2.4 GHz
BladeRF 2.0 micro case	1	20 US dollar	PolyCarbonate case for the BladeRF micro A4
RFSpace TSA600	2	74 US dollar	UWB tapered slot antenna Frequency range: 600 MHz to 6 GHz Gain: 7 to 11 dBi
ZFDC-20-50-S+	1	97.53 Euro	Directional Coupler Frequency range: 20 MHz to 2 GHz Coupling Factor: 19.5 +/- 0.5 dB
BECEN Attenuator	1	11.5 Euro	-20 dB attenuator
SMA cables	4	8.5 Euro	Male to male 30 cm

Software

List of Dependencies

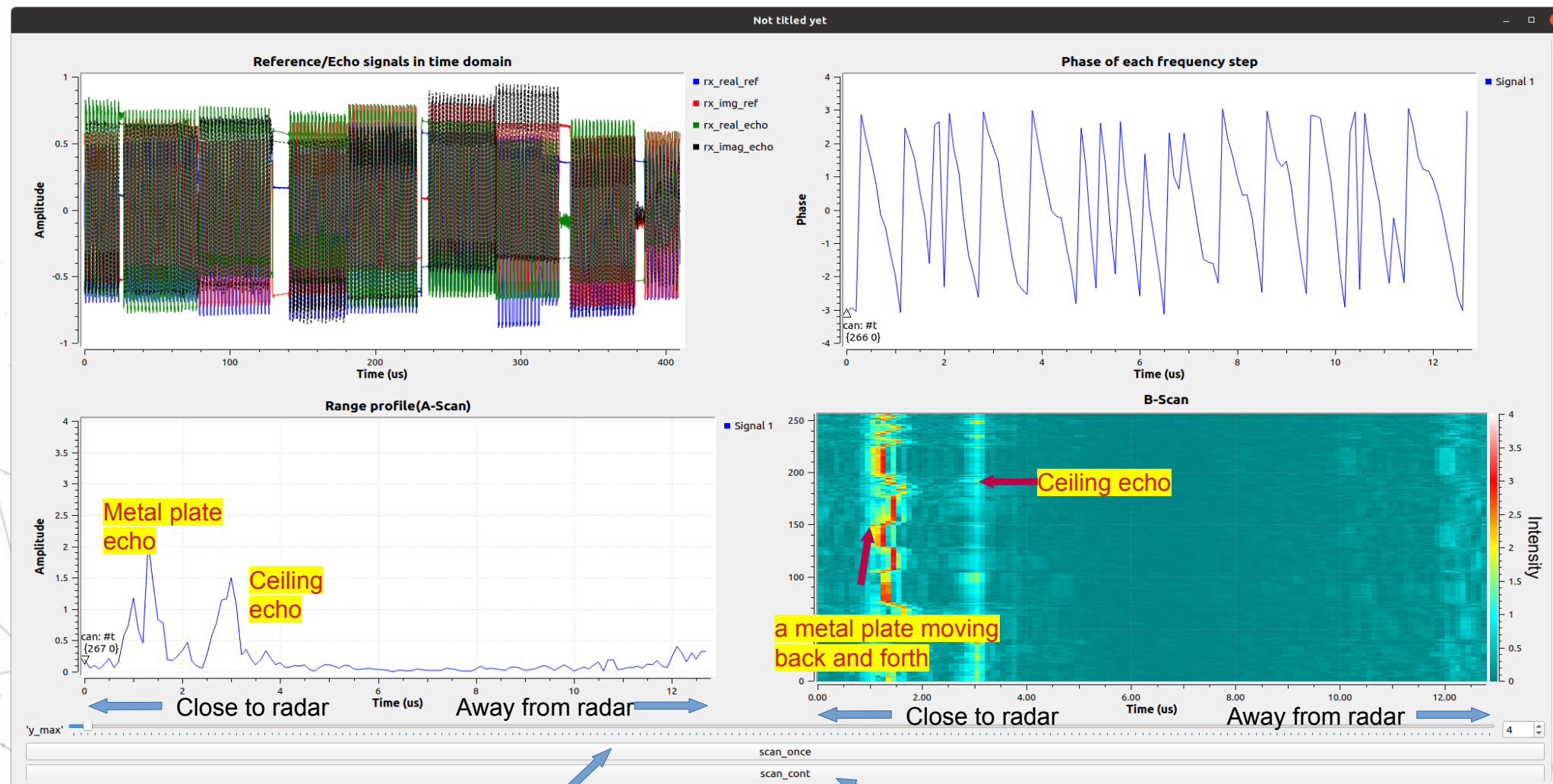
Library	Version	Licensing
libbladeRF	2.5.0	LGPLv2.1
GNURadio	3.10	GPLv3
volk	2.4	GPLv3

Library	Version	Licensing
Python3	3.11	PSF
numpy	1.17.4	MIT
scipy	1.3.3	MIT

Software components

Module	Status	Language	DIR	Functions
bladerf_device.cc	Developed	c++	lib/	Base class to control bladeRF to transmit pulse and receive echo signals
sfcw_radar_mimo_cc_impl.cc	Developed	c++	lib/	Sfcw radar implementation using the second TX channel for reference signals
sfcw_radar_miso_cc_impl.cc	Developed	c++	lib/	Sfcw radar implementation using direction coupler for reference signals
RawSamplesSink.py	Developed	python	python/sfcwRadar/	Record raw radar echos
rangeProfileSink.py	Developed	python	python/sfcwRadar	Record range profile and send to the remote server
ClutterRemover.py	Developed	python	script/	Remove ground clutter
MatchedFilter.py	Developed	python	python/sfcwRadar/	Matched filter implementation to extract channel response when using chirp waveform
Calibration.py	Developed	python	python/sfcwRadar/	Calibration algorithm to address the internal coupling
GetRangeProfile.py	Developed	python	script/	Signal processing pipeline from radar echos to range profile (A-scan)
sfcwRadarGui	Developed	python+grc	gui/	The graph user interface of the ground penetrating radar. Use it for data collection or onsite survey

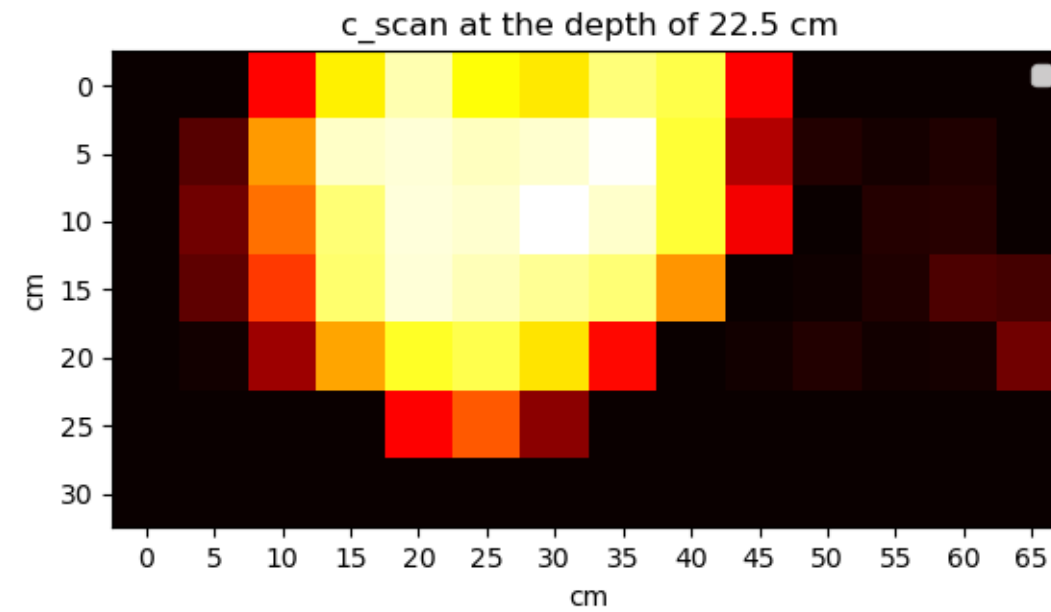
A-scan and B-scan example



Click this button
to scan just
once

Scan contentiously

C-Scan example



Experiments

